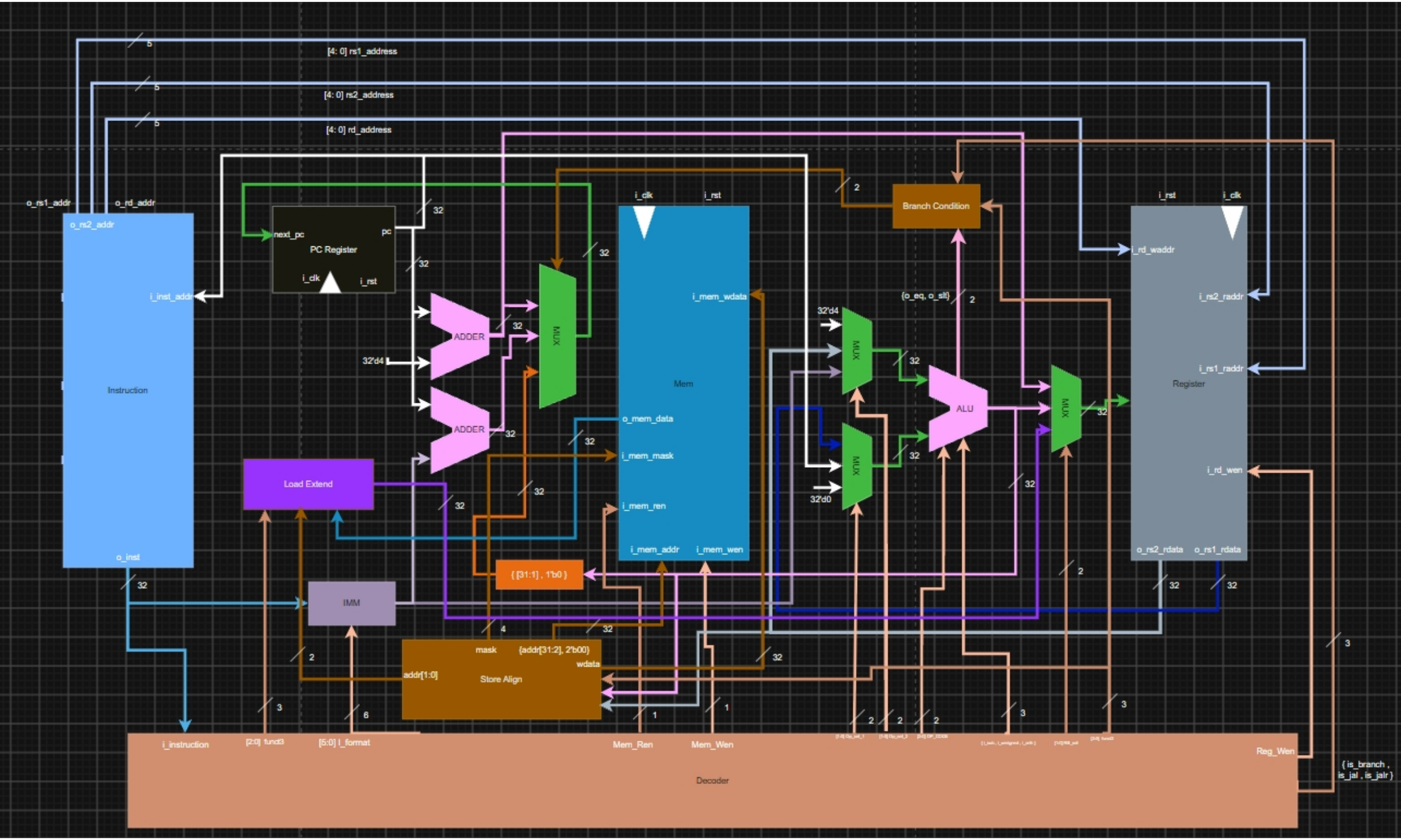

[4: 0] rs1_address
[4: 0] rs2_address
[4: 0] rd_address
o_rs1_addr
o_rd_addr
o_rs2_addr
Instruction
i_inst_addr
o_inst
PC Register
next_pc
pc
i_clk
i_rst
ADDER
ADDER
32'd4
MUX
Load Extend
IMM
{ [31:1] , 1'b0 }
Mem
i_mem_wdata
o_mem_data
i_mem_mask
i_mem_ren
i_mem_addr
i_mem_wen
Store Align
mask
{addr[31:2], 2'b00}
addr[1:0]
wdata
Branch Condition
{o_eq, o_slt}
MUX
MUX
MUX
32'd4
32'd0
ALU
Register
i_rd_waddr
i_rs2_raddr
i_rs1_raddr
i_rd_wen
o_rs2_rdata
o_rs1_rdata
i_instruction
[2:0] funct3
[5:0] I_format
Mem_Ren
Mem_Wen
Reg_Wen
Decoder
{ is_branch , is_jal , is_jalr }